

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
6	(a)		Attempt at starting from Faraday's law e.g. $\frac{BA}{t}$ (1) Clear explanation leading to Bvd – minimum $\frac{BA}{t} = \frac{Bxd}{t} = Bdv$ OR area in 1 s is dv (1)	2			2	1	
	(b)	(i)	Flux in or right pd (in 1 s) = B_2vd (1) Flux out or left pd (in 1 s) = B_1vd (1) Net flux change or pd around the loop (per s) = $B_2vd - B_1vd$ (1) Alternative: $V = \frac{d}{dt}(BA) = \frac{AdB}{dt}$ (1) $V = \frac{AdB}{dx} \frac{dx}{dt}$ (1) $V = d^2 \left(\frac{B_2 - B_1}{d} \right) v$ QED (1) Award 2 marks for: pd on right = B_2vd pd on left = B_1vd Hence $B_2vd - B_1vd$ Award 1 mark for: It depends on the change in flux / B Hence ΔBvd	1	1 1		3	1	
		(ii)	$B_2 - B_1 = 1.20 \times 0.038$ (1) Answer = 15.25 m[V] or $1.2 \times 0.038 \times 8.8 \times 0.038$ seen (1)		2		2	2	

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
		(iii)	Length = $4 \times 3.8\text{cm}$ (1) Area = $2.2\text{mm} \times 2\text{mm}$ (1) Resistivity equation used ($0.55\text{m}\Omega$) (1) $I = \frac{V}{R}$ used (1) Correct answer = 28 [A] (1) ecf 56 A and 111 A award 4 marks	1 1	1 1		5	3	
			Question 6 total	5	7	0	12	7	0